

1. Abed, E.H., and J-H. Fu, "Local stabilization and bifurcation control, I. Hopf bifurcation," *Systems and Control Letters* **7**(1986): 11-17.
2. Abraham, R., and J.E. Marsden, *Foundations of Mechanics, second edition*, Benjamin Cummings, Reading, 1978.
3. Adamjan, V.M., D.Z. Arov, and M.G. Krein, "Analytic properties of Schmidt pairs for a Hankel operator and the generalized Schur-Takagi problem," *Math. USSR Sbornik* **15**(1971): 31-73.
4. Aeyels, D., "Stabilization of a class of nonlinear systems by a smooth feedback control," *Systems and Control Letters* **5**(1985): 289-294.
5. Aeyels, D., "Generic observability of differentiable systems," *SIAM J. Control & Opt.* **19**(1981): 595-603.
6. Aeyels, D., "On the number of samples necessary to achieve observability," *Systems and Control Letters* **1**(1981): 92-94.
7. Aeyels, D., "Global observability of Morse-Smale vector fields," *J. Diff. Eqs.* **45**(1982): 1-15.
8. Aeyels, D., and M. Szafranski, "Comments on the stabilizability of the angular velocity of a rigid body," *Systems and Control Letters* **10**(1988): 35-39.
9. Ahmed, N.U., *Elements of Finite-Dimensional Systems and Control Theory*, Wiley, New York, 1988.
10. Akaike, H., "Stochastic theory of minimal realization," *IEEE Trans. Autom. Control* **AC-19**(1974): 667-674.
11. Akaike, H., "Markovian representation of stochastic processes by canonical variables," *SIAM J. Control & Opt.* **13**(1975):162-173.
12. Albertini, F., and P. Dai Pra, "Forward accessibility for recurrent neural networks," *IEEE Trans. Automat. Control* **40** (1995): 1962-1968.
13. Albertini, F., and E.D. Sontag, "For neural networks, function determines form," *Neural Networks* **6**(1993): 975-990.
14. Albertini, F., and E.D. Sontag, "State observability in recurrent neural networks," *Systems & Control Letters* **22**(1994): 235-244.
15. Alur, R., T.A. Henzinger, and E.D. Sontag (eds.), *Hybrid Systems III. Verification and Control*, Springer Verlag, Berlin, 1996.
16. Anderson, B.D.O., and J.B. Moore, *Linear Optimal Control*, Prentice-Hall, Englewood Cliffs, 1971.
17. Antsaklis, P.J., and A.N. Michel, *Linear Systems*, McGraw Hill, New York, 1997.
18. Antoulas, A.C., "On recursiveness and related topics in linear systems," *IEEE Trans. Autom. Control* **AC-31**(1986): 1121-1135.
19. Antoulas, A.C., and B.D.O. Anderson, "On the scalar rational interpolation problem," *IMA J. Math. Control Infor.* **3**(1986):61-88.
20. Aoki, M., *Optimal Control and System Theory in Dynamic Economic Analysis*, North Holland, New York, 1976.

21. Arbib, M.A., "A characterization of multilinear systems," *IEEE Trans. Autom. Control* **14**(1969): 699-702.
22. Arbib, M.A., and E. Manes, "Machines in a category: an expository introduction," *SIAM Review* **57**(1974): 163-192.
23. Artstein, Z., "Stabilization with relaxed controls," *Nonl. Anal., TMA* **7**(1983): 1163-1173.
24. Astrom, K.J., *Introduction to Stochastic Control Theory*, Academic Press, New York, 1970.
25. Athans, M., and P.L. Falb, *Optimal Control: An Introduction to the Theory and its Applications*, McGraw-Hill, New York, 1966.
26. Aubin, J.-P., and A. Cellina, *Differential Inclusions: Set-Valued Maps and Viability Theory*, Springer-Verlag, Berlin, 1984.
27. Baccioti, A., and P. Boieri, "Linear stabilizability of planar nonlinear systems," *Math. Control Signals Systems* **3**(1990): 183-193.
28. Baillieul, J., "Controllability and observability of polynomial dynamical systems," *Nonl. Anal., TMA* **5**(1981): 543-552.
29. Baillieul, J., R.W. Brockett, and R.B. Washburn, "Chaotic motion in nonlinear feedback systems," *IEEE Trans. Circuits and Systems* **CAS-27**(1980): 990-997.
30. Baillieul, J., S.S. Sastry, and H.J. Sussmann (eds.), *Essays on Mathematical Robotics*, Springer-Verlag, New York, 1998.
31. Banks, S.P., *Control Systems Engineering: Modelling and Simulation, Control Theory, and Microprocessor Implementation*, Prentice-Hall, Englewood Cliffs, 1986.
32. Banks, S.P., "Stabilizability of finite- and infinite-dimensional bilinear systems," *IMA J. of Math. Control and Inform.* **3**(1986): 255-271.
33. Barmish, B.R., M.J. Corless, and G. Leitmann, "A new class of stabilizing controllers for uncertain dynamical systems," *SIAM J. Control & Opt.* **21**(1983): 246-255.
34. Barnett, S., *Polynomials and Linear Control Systems*, Marcel Dekker, New York, 1983.
35. Barnett, S., *Matrices in Control Theory*, Krieger, Malabar, 1984.
36. Barnett, S., *Introduction to Mathematical Control Theory*, Oxford University Press, New York, 1985.
37. Bartlett, A.C., C.V. Hollot, and H. Lin, "Root locations of an entire polytope of polynomials: It suffices to check the edges," *Math. Control Signals Systems* **1**(1988): 61-71.
38. Bartosiewicz, Z., "Minimal polynomial realizations," *Math. Control Signals Systems* **1**(1988): 227-231.
39. Basile, G., and G. Marro, "Controlled and conditioned invariant subspaces in linear system theory," *J. Optimiz. Theo. & Appl.* **3**(1969): 306-315.
40. Belevitch, V., *Classical Network Theory*, Holden-Day, San Francisco, 1968.

41. Bellman, R., *Dynamic Programming*, Princeton University Press, Princeton, 1957.
42. Bellman, R.E., *Introduction to the Mathematical Theory of Control Processes*, Academic Press, New York, 1967.
43. Bellman, R.E., and R. Kalaba (eds.), *Selected Papers on Mathematical Trends in Control Theory*, Dover, New York, 1964.
44. Bellman, R.E., and R. Kalaba, *Dynamic Programming and Modern Control Theory*, Academic Press, New York, 1965.
45. Bensoussan, A., E.G. Hurst, and B. Naslund, *Management Applications of Modern Control Theory*, Elsevier, New York, 1974.
46. Benveniste, A., and P. Le Guernic, "Hybrid dynamical systems theory and the SIGNAL language," *IEEE Trans. Autom. Control* **AC-35**(1990): 535-546.
47. Berger, M.S., and M.S. Berger, *Perspectives in Nonlinearity: An Introduction to Nonlinear Analysis*, Benjamin, New York, 1968.
48. Berkovitz, L.D., *Optimal Control Theory*, Springer-Verlag, New York, 1974.
49. Bhatia, N.P., and G.P. Szegö, *Stability Theory in Dynamical Systems*, Springer-Verlag, Berlin, 1970.
50. Boltyansky, V.G., "Sufficient conditions for optimality and the justification of the dynamic programming method," *SIAM J. Control & Opt.* **4**(1966): 326-361.
51. Bonnard, B., "Controlabilité des systèmes non linéaires," *C.R. Acad. Sci. Paris, Serie I*, **292**(1981): 535-537.
52. Booth, T.L., *Sequential Machines and Automata Theory*, Wiley, New York, 1967.
53. Boothby, W. M., *An Introduction to Differentiable Manifolds and Riemannian Geometry*, Academic Press, New York, 1975.
54. Boothby, W.M., and R. Marino, "Feedback stabilization of planar nonlinear systems," *Systems and Control Letters* **12**(1989): 87-92.
55. Boothby, W.M., and E.N. Wilson, "Determination of the transitivity of bilinear systems," *SIAM J. Control & Opt.* **17**(1979): 212-221.
56. Bose, N.K., "Problems and progress in multidimensional systems theory," *Proceedings IEEE* **65**(1976): 824-840.
57. Bose, N.K., *Multidimensional Systems Theory*, Reidel, Dordrecht, 1985.
58. Boyd, S.P., and C.H. Barratt, *Linear Controller Design*, Prentice Hall, Englewood Cliffs, 1991.
59. Boyd, S., V. Balakrishnan, and P. Kabamba, "On computing the H_∞ norm of a transfer matrix," *Math. Control Signals Systems* **2** (1989): 207-219.
60. Brammer, R., "Controllability in linear autonomous systems with positive controllers," *SIAM J. Control* **10**(1972): 339-353.
61. Brewer, J.W., J.W. Bunce, and F.S. Van Vleck, *Linear Systems over Commutative Rings*, Marcel Dekker, New York, 1986.

62. Brewer, J.W., and L. Klinger, "Pole shifting for families of systems: the dimension one case," *Math. Control Signals Systems* **1**(1988): 285-292.
63. Brockett, R.W., "The status of stability theory for deterministic systems," *IEEE Trans. Autom. Control* **AC-11**(1966): 596-606.
64. Brockett, R.W., "Structural properties of the equilibrium solutions of Riccati equations," in *Lect. Notes in Math., vol. 132* (A.V. Balakrishnan et al., eds.), Springer-Verlag, Heidelberg, 1970, pp. 61-69.
65. Brockett, R.W., *Finite Dimensional Linear Systems*, Wiley, New York, 1970.
66. Brockett, R.W., "On the algebraic structure of bilinear systems," in *Theory and Applications of Variable Structure Systems* (R. Mohler and A. Ruberti, eds.), Academic Press, New York, 1972, pp. 153-168.
67. Brockett, R.W., "System theory on group manifolds and coset spaces," *SIAM J. Control* **10**(1972): 265-284.
68. Brockett, R.W., "Lie theory and control systems defined on spheres," *SIAM J. Control & Opt.* **25**(1973): 213-225.
69. Brockett, R.W., "Volterra series and geometric control theory," *Automatica* **12**(1976): 167-176.
70. Brockett, R.W., "Nonlinear systems and differential geometry," *Proc. IEEE* **64**(1976): 61-72.
71. Brockett, R.W., "Feedback invariants for nonlinear systems," *Proc. IFAC Congress*, Helsinki, 1978.
72. Brockett, R.W., "Asymptotic stability and feedback stabilization," in *Differential Geometric Control Theory* (R.W. Brockett, R.S. Millman, and H.J. Sussmann, eds.), Birkhauser, Boston, 1983, pp. 181-191.
73. Brockett, R.W., R.S. Millman, and H.J. Sussmann (eds.), *Differential Geometric Control Theory*, Birkhauser, Boston, 1983.
74. Brockett, R.W., and J.L. Willems, "Discretized partial differential equations: Examples of control systems defined on modules," *Automatica* **10**(1974): 507-515.
75. Brockett, R.W., and A.S. Willsky, "Finite group homomorphic sequential systems," *IEEE Trans. Autom. Control* **AC-17**(1972): 483-490.
76. Brogan, W.L., *Modern Control Theory*, Prentice-Hall, Englewood Cliffs, 1985.
77. Brunovsky, P., "A classification of linear controllable systems," *Kybernetika* **6**(3)(1970): 173-188.
78. Bryson, A.E., and Y-C. Ho, *Applied Optimal Control*, Wiley, New York, 1969.
79. Bumby, R., E.D. Sontag, H.J. Sussmann, and W. Vasconcelos, "Remarks on the pole-shifting problem over rings," *J. Pure Appl. Algebra* **20**(1981): 113-127.
80. Byrnes, C.I., and A. Isidori, "New results and counterexamples in nonlinear feedback stabilization," *Systems and Control Letters* **12**(1989): 437-442.

81. Byrnes, C.I., "On the control of certain infinite-dimensional systems by algebro-geometric techniques," *Amer. J. Math.* **100**(1978): 1333-1381.
82. Campbell, S.L., *Singular Systems of Differential Equations, I and II*, Pitman, Marshfield, 1980, 1982.
83. Campbell, S.L., "Control problem structure and the numerical solution of linear singular systems," *Math. Control Signals Systems* **1**(1988): 73-87.
84. Canudas de Wit, C.A., *Adaptive Control for Partially Known Systems: Theory and Applications*, Elsevier, New York, 1988.
85. Casdagli, M., "Nonlinear prediction of chaotic time series," *Physica D* **35**(1989): 335-356.
86. Catlin, D.E., *Estimation, Control, and the Discrete Kalman Filter*, Springer-Verlag, New York, 1989.
87. Celle, F., J.P. Gauthier, D. Kazakos, and G. Sallet, "Synthesis of nonlinear observers: A harmonic analysis approach," *Math. Systems Theory (Special Issue on Mathematical Control Theory)* **22**(1989): 291-322.
88. Chen, C.T., *Introduction to Linear System Theory*, Holt, Reinhart and Winston, New York, 1970.
89. Chen, K.T., "Iterated path integrals," *Bulletin AMS* **83**(1977): 831-879.
90. Cho, H., and S.I. Marcus, "On supremal languages of classes of sublanguages that arise in supervisor synthesis problems with partial observation," *Math. Control Signals Systems* **2**(1989): 47-69.
91. Chow, W.L., "Über systeme von linearen partiellen differentialgleichungen erster ordnung," *Math. Ann.* **117**(1939): 98-105.
92. Clarke, F.H., Yu.S. Ledyae, E.D. Sontag, and A.I. Subbotin, "Asymptotic controllability implies feedback stabilization," *IEEE Trans. Autom. Control* **42**(1997): 1394-1407.
93. Clarke, F.H., Yu.S. Ledyae, R.J. Stern, and P. Wolenski, *Nonsmooth Analysis and Control Theory*, Springer-Verlag, New York, 1997.
94. Cobb, J.D., "Feedback and pole placement in descriptor variable systems," *Intern. J. Control* **33**(1981): 1135-1146.
95. Conte, G., and A.M. Perdon, "An algebraic notion of zeros for systems over rings," in *Mathematical Theory of Networks and Systems*, (P.A. Fuhrmann, ed.), Springer-Verlag, Berlin, 1984, pp. 323-330.
96. Constanza, V., B.W. Dickinson, and E.F. Johnson, "Universal approximations of discrete-time control systems over finite time," *IEEE Trans. Autom. Control* **AC-28**(1983): 439-452.
97. Conway, J.B., *Regular Algebra and Finite Machines*, Chapman and Hall, London, 1971.
98. Conway, J.B., *A Course in Functional Analysis, Second Edition*, Springer-Verlag, New York, 1990.
99. Coron, J-M., "Global asymptotic stabilization for controllable systems without drift," *Math of Control, Signals, and Systems* **5**(1992): 295-312.

100. Coron, J-M., "Stabilization in finite time of locally controllable systems by means of continuous time-varying feedback laws," *SIAM J. Control and Opt.* **33**(1995): 804-833.
101. Coron, J-M., L. Praly, and A. Teel, "Feedback stabilization of nonlinear systems: sufficient conditions and Lyapunov and input-output techniques," in *Trends in Control* (A. Isidori, ed.), Springer-Verlag, Berlin, 1995, pp. 293-348.
102. Crandall, M.G., L.C. Evans, and P.L. Lions, "Some properties of viscosity solutions of Hamilton-Jacobi equations," *Trans. AMS* **282**(1984): 487--502.
103. Craven, B.D., *Mathematical Programming and Control Theory*, Wiley, New York, 1978.
104. Crouch, P.E., "Dynamical realizations of finite Volterra series," Ph.D. Thesis, Harvard, 1977.
105. Crouch, P.E., "Spacecraft attitude control and stabilization: applications of geometric control theory," *IEEE Trans. Autom. Control* **29**(1984): 321-333.
106. Crouch, P.E., and A.J. van der Schaft, *Variational and Hamiltonian Control Systems*, Springer-Verlag, New York, 1987.
107. Curtain, R., and A.J. Pritchard, *Infinite Dimensional Linear Systems Theory*, Springer-Verlag, Berlin, 1978.
108. Davis, M.H.A., *Linear Estimation and Stochastic Control*, Chapman and Hall, London, 1977.
109. Dayawansa, W.P., and C.F. Martin, "Asymptotic stabilization of two dimensional real-analytic systems," *Systems Control Lett.* **12**(1989): 205-211.
110. de Jong, L.S., *Numerical Aspects of Realization Algorithms in Linear Systems Theory*, Ph.D. Thesis, Univ. of Eindhoven, Mathematics Department, 1975.
111. Deimling, K., *Multivalued Differential Equations*, de Gruyter, Berlin, 1992.
112. Delchamps, D.F., "Analytic stabilization and the algebraic Riccati equation," in *Proc. IEEE Conf. Dec. and Control, Dec. 1983*, IEEE Publications, pp. 1396-1401.
113. Delfour, M.C., C. McCalla, and S.K. Mitter, "Stability and the infinite-time quadratic cost problem for linear hereditary differential systems," *SIAM J. Control* **13**(1975): 48-88.
114. Desoer, C.A., *Notes for a Second Course on Linear Systems*, Van Nostrand, New York, 1970.
115. Desoer, C.A., and M. Vidyasagar, *Feedback Synthesis: Input-Output Properties*, Academic Press, New York, 1975.
116. Desoer, C.A., and M.G. Kabuli, "Nonlinear plants, factorizations, and stable feedback systems," in *Proc. IEEE Conf. Decision and Control*, Los Angeles, Dec 1987, pp. 155-156.

117. Dickinson, B.W., "On the fundamental theorem of linear state variable feedback," *IEEE Trans. Autom. Contr.* **AC-19**(1974):577-579.
118. Dickinson, B.W., "A new characterization of canonical convolutional encoders," *IEEE Trans. Inform. Theory* **IT-20**(1976): 352-354.
119. Dickinson, B.W., "The classification of linear state variable control laws," *SIAM J. Control & Opt.* **14**(1976): 467-477.
120. Dickinson, B.W., and E.D. Sontag, "Dynamical realizations of sufficient sequences," *IEEE Trans. Inform. Th.* **IT-31**(1985):670-676.
121. Dieudonné, J., *Foundations of Modern Analysis*, Academic Press, New York, 1960.
122. Dolezal, V., *Monotone Operators and Applications in Control and Network Theory*, Elsevier, New York, 1979.
123. Dorato, P. (ed.), *Robust Control*, IEEE Press, New York, 1987.
124. Doyle, J.C., K. Glover, P.P. Khargonekar, and B.A. Francis, "State-space solutions to the standard H_2 and H_∞ control problems," *IEEE Trans. Autom. Control* **AC-34**(1989): 831-846.
125. Eilenberg, S., *Automata, Languages, and Machines, Volume A*, Academic Press, New York, 1974.
126. El-Assoudi, R., and J.P. Gauthier, "Controllability of right-invariant systems on real simple Lie groups of type F_4 , G_2 , C_n , and B_n ," *Math. Control Signals Systems* **1**(1988): 293-301
127. Elliott, D.L., "A consequence of controllability," *J. Diff. Eqs.* **10** (1971): 364-370.
128. Emre, E., "On necessary and sufficient conditions for regulation of linear systems over rings," *SIAM J. Control & Opt.* **20**(1982): 155-160.
129. Emre, E., and P.P. Khargonekar, "Regulation of split linear systems over rings: coefficient-assignment and observers," *IEEE Trans. on Autom. Control* **AC-27**(1982): 104-113.
130. Farmer, J.D., and J.J. Sidorowich, "Exploiting chaos to predict the future and reduce noise," in *Evolution, Learning, and Cognition* (Y.C. Lee, ed.) World Scientific Press, 1988, p. 277ff.
131. Faurre, P., "Stochastic realization algorithms," in *System Identification - Advances and Case Studies* (R.K. Mehra and D.G. Lainiotis, eds.), Academic Press, New York, 1976, pp. 1-25.
132. Faurre, P., M. Clerget, and F. Germain, *Operateurs Rationnels Positifs*, Dunod, Paris, 1979.
133. Feintuch, A., and R. Sacks, *System Theory. A Hilbert Space Approach*, Academic Press, New York, 1982.
134. Fleming, W.H., and R.W. Rishel, *Deterministic and Stochastic Optimal Control*, Springer-Verlag, New York, 1975.
135. Fleming, W.H., et. al., *Future Directions in Control Theory: A Mathematical Perspective*, SIAM Publications, Philadelphia, 1988.
136. Fliess, M., "Sur certaines familles de séries formelles," Thèse de Doctorat d'Etat, Université Paris VII, 1972.

137. Fliess, M., "Matrices de Hankel," *J. Math Pures et. Appl.* **53** (1974): 197-224.
138. Fliess, M., "Sur la realization des systemes dynamiques bilineaires," *C.R. Acad. Sc. Paris* **A277**(1973): 243-247.
139. Fliess, M., "The unobservability ideal for nonlinear systems," *IEEE Trans. Autom. Control* **AC-26**(1981): 592-593.
140. Fliess, M., "Fonctionnelles causales non linéaires et indéterminées non commutatives," *Bull. Soc. Math. France* **109**(1981): 3-40.
141. Fliess, M., "Quelques définitions de la théorie des systèmes à la lumière des corps différentiels," *C.R.Acad. Sc. Paris* **I**(1987): 91-93.
142. Fliess, M., and I. Kupka, "A finiteness criterion for nonlinear input-output differential systems," *SIAM J.Control & Opt.* **21**(1983): 721-728.
143. Fornasini, E., and G. Marchesini, "Algebraic realization theory of bilinear discrete-time input/output maps," *J.Franklin Inst.* **301**(1976): 143-159.
144. Fortmann, T.E., and K.L.Hitz, *An Introduction to Linear Control Systems*, Marcel Dekker, New York, 1977.
145. Francis, B.A., *A Course in H_∞ Control Theory*, Springer-Verlag, New York, 1987.
146. Fruechte, R.D., *Application of Control Theory in the Automotive Industry*, Interscience Enterprises, Geneve, 1983.
147. Fuhrmann, P.A., *Linear Systems and Operators in Hilbert Space*, McGraw-Hill, New York, 1981.
148. Fujiwara, H., "Computational complexity of controllability/observability problems for combinatorial circuits," in *Proc. 18th IEEE Int. Symp. on Fault Tolerant Computing*, IEEE Publ., 1988, pp. 64-69.
149. Fuller, A.T., "The early development of control theory, parts I and II," *J. Dynamic Systems, Measurement, and Control* **98**(1976): 109-118 and 224-235.
150. Fuller, A.T., "In the large stability of relay and saturated control systems with linear controllers," *Int. J. Control*, **10**(1969): 457-480.
151. Gamkrelidze, R.V., *Principles of Optimal Control Theory*, Plenum Press, New York, 1978.
152. Gantmacher, F.R., *Theory of Matrices* (English transl.), Chelsea, New York, 1959.
153. Gelb, A., et.al., *Optimal Estimation*, MIT Press, Cambridge, MA, 1974.
154. Gibson, J.A., and T.T. Ha, "Further to the preservation of controllability under sampling," *Int. J. Control* **31**(1980): 1013-1026.
155. Gilbert, E., "Controllability and observability in multivariable control systems," *SIAM J. Control* **1**(1963): 128-151.
156. Ginsburg, S., *An Introduction to Mathematical Machine Theory*, Addison-Wesley, Reading, MA, 1962.
157. Glass, M., "Exponential stability revisited," *Int. J. Control* **46**(1987): 1505-1510.

158. Gopal, M., *Modern Control System Theory*, Wiley, New York, 1984.
159. Gopinath, B., "On the control of multiple input-output systems," *Bell Systems Tech. J.*, March 1971.
160. Grasse, K.A., "Perturbations of nonlinear controllable systems," *SIAM J. Control & Opt.* **19**(1981): 203-220.
161. Grasse, K.A., "Nonlinear perturbations of control-semilinear control systems," *SIAM J. Control & Opt.* **20**(1982): 311-327.
162. Grasse, K.A., "On accessibility and normal accessibility: the openness of controllability in the fine C^0 topology," *J. Diff. Eqs.* **53**(1984): 387-414.
163. Grasse, K.A., and H.J. Sussmann, "Global controllability by nice controls," in *Nonlinear Controllability and Optimal Control* (H.J. Sussmann, ed.), Dekker, New York, 1990, pp. 33-79.
164. Grasselli, O.M., and A. Isidori, "An existence theorem for observers of bilinear systems," *IEEE Trans. Autom. Control* **26**(1981): 1299-1301.
165. Grimble, M.J., and M.A. Johnson, *Optimal Control and Stochastic Estimation: Theory and Applications*, Wiley, New York, 1988.
166. Grizzle, J.W., "Controlled invariance for discrete-time nonlinear systems with an application to the decoupling problem," *IEEE Trans. Autom. Control* **AC-30**(1985): 868-874.
167. Grizzle, J.W., and A. Isidori, "Block noninteracting control with stability via static state feedback," *Math. Control Signals Systems* **2**(1989): 315-341.
168. Grodins, F.S., *Control Theory and Biological Systems*, Columbia University Press, New York, 1963.
169. Hahn, W., *Stability of Motion*, Springer-Verlag, New York, 1967.
170. Hájek, O., *Control Theory in the Plane*, Springer-Verlag, Berlin, 1991.
171. Hammer, J., "Fraction representations of nonlinear systems: a simplified approach," *Int. J. Control* **46**(1987): 455-472.
172. Hammer, J., "On nonlinear systems, additive feedback, and rationality," *Int. J. Control* **40**(1984): 953-969.
173. Harris, C.J., and S.A. Billings, *Self-Tuning and Adaptive Control: Theory and Applications*, P. Peregrinus, London, 1985.
174. Harvey, C.A., and E.B. Lee, "On stability with linear feedback control," Honeywell Corp. Memo Number 8150 (to C.R. Stone), January 8, 1963.
175. Hautus, M.L.J., "Controllability and observability conditions for linear autonomous systems," *Ned. Akad. Wetenschappen, Proc. Ser. A* **72** (1969): 443-448.
176. Hautus, M.L.J., "Stabilization, controllability, and observability for linear autonomous systems," *Ned. Akad. Wetenschappen, Proc. Ser. A* **73**(1970): 448-455.
177. Hautus, M.L.J., "A simple proof of Heymann's Lemma," Memo COSOR 76-17, Department of Mathematics, Eindhoven Univ. of Technology, Nov. 76.

178. Hautus, M.L.J., “ (A, B) -invariant and stabilizability subspaces, a frequency domain description,” *Automatica* **16**(1980): 703-707.
179. Hautus, M.L.J., and E.D. Sontag, “New results on pole-shifting for parametrized families of systems”, *J.Pure Appl. Algebra* **40**(1986): 229-244.
180. Hazewinkel, M., and A-M. Perdon, “On families of systems: pointwise-local-global isomorphism problems,” *Int.J. Control* **33**(1981): 713-726.
181. Helmke, U., “Linear dynamical systems and instantons in Yang-Mills theory,” *IMA J.Math. Control and Information* **3**(1986): 151-166.
182. Hermann, R., “On the accessibility problem in control theory,” in *Int. Symp. Diff. Eqs. and Mechanics*, Academic Press, New York, 1973.
183. Hermann, R., *Geometric Structure Theory of Systems-Control, Theory and Physics*, Math. Sci. Press, Brookline, MA, 1974.
184. Hermann, R., *Cartanian Geometry, Nonlinear Waves, and Control Theory*, Math. Sci. Press, Brookline, MA, 1979.
185. Hermann, R., and A.J. Krener, “Nonlinear controllability and observability,” *IEEE Trans. Autom. Control* **22**(1977): 728-740.
186. Hermes, H., “On a stabilizing feedback attitude control,” *J. Optim. Theory and Appl.* **31**(1980): 373-384.
187. Hermes, H., “On the synthesis of a stabilizing feedback control via Lie algebraic methods,” *SIAM J. Control & Opt.* **18**(1980):352-361.
188. Hermes, H., and J.P. LaSalle, *Functional Analysis and Time-Optimal Control*, Academic Press, New York, 1969.
189. Hestenes, M.R., *Calculus of Variations and Optimal Control Theory*, R. E. Kriegerz, Huntington, New York, 1980.
190. Hewitt, E., and K. Stromberg, *Real and Abstract Analysis*, Springer-Verlag, New York, 1965.
191. Heymann, M., “Comments on ‘Pole assignment in multi-input controllable linear systems’,” *IEEE Trans. Autom. Control* **AC-13** (1968): 748-749.
192. Hinrichsen, D., and J. O’Halloran, “A complete characterization of orbit closures of controllable singular systems under restricted system equivalence,” *SIAM J. Control & Opt.* **28**(1990): 602-623.
193. Hirschorn, R.M., “Controllability in nonlinear systems,” *J. Diff. Eqs.* **19**(1975): 46-61.
194. Hirschorn, R.M., “Invertibility of nonlinear control systems,” *SIAM J. Control & Opt.* **17**(1979): 289-297.
195. Hirschorn, R.M., “ (A, B) -invariant distributions and disturbance decoupling in nonlinear systems,” *SIAM J. Control & Opt.* **19** (1981): 1-19.
196. Hsu, F.H., *On Reachable Sets*, Ph.D. Thesis, Case Western Reserve University, Cleveland, 1973.
197. Hunt, L.R., R. Su, and G. Meyer, “Design for multi-input nonlinear systems,” in *Differential Geometric Control theory* (R.W.Brockett, R.S.Millman, and H.J.Sussmann, eds.), Birkhauser, Boston, 1983.

198. Isidori, A., "Direct construction of minimal bilinear realizations from nonlinear input/output maps," *IEEE Trans. Autom. Control* **18**(1973): 626-631.
199. Isidori, A., *Nonlinear Control Systems, Third Edition*, Springer-Verlag, London, 1995.
200. Isidori, A., A.J. Krener, C. Gori-Giorgi, and S. Monaco, "Nonlinear decoupling via feedback: A differential geometric approach," *IEEE Trans. Autom. Control* **26**(1981): 331-345.
201. Jacobson, C.A., and C.N. Nett, "Linear state-space systems in infinite-dimensional space: The role and characterization of joint stabilizability/detectability," *IEEE Trans. Autom. Control* **33**(1988): 541-549.
202. Jacobson, D.H., *Extensions of Linear-Quadratic Control, Optimization and Matrix Theory*, Academic Press, New York, 1977.
203. Jakubczyk, B., "Invertible realizations of nonlinear discrete time systems," in *Proc. Princeton Conf. Inf. Sc. and Sys.*(1980): 235-239.
204. Jakubczyk, B., "Poisson structures and relations on vector fields and their Hamiltonians," *Bull. Polish Acad. Sci.* **34**(1986): 713-721.
205. Jakubczyk, B., "Feedback linearization of discrete-time systems," *Systems and Control Letters* **9**(1987): 441-416.
206. Jakubczyk, B., and D. Normand-Cyrot, "Orbites de pseudo-groupes de difféomorphismes et commandabilité des systèmes non linéaires en temps discret," *C.R. Acad. Sc. Paris* **298-I**(1984): 257-260.
207. Jakubczyk, B., and W. Respondek, "On linearization of control systems," *Bull. Acad. Pol. Sci., Ser. Sci. Math. Astr. Phys.* **28**(1980): 517-522.
208. Jakubczyk, B., and E.D. Sontag, "Controllability of nonlinear discrete-time systems: A Lie-algebraic approach," *SIAM J. Control & Opt.* **28**(1990): 1-33.
209. James, M.R., "Asymptotic nonlinear filtering and large deviations with application to observer design," Systems Research Center Report TR-88-28, University of Maryland, 1988. (Ph. D. Dissertation.)
210. Jurdjevic, V., and J.P. Quinn, "Controllability and stability," *J. Diff. Eqs.* **28**(1978): 381-389.
211. Jurdjevic, V., and H.J. Sussmann, "Control systems on Lie groups," *J. Diff. Eqs.* **12**(1972): 313-329.
212. Kailath, T., *Linear Systems*, Prentice Hall, Englewood Cliffs, 1980.
213. Kalman, R.E., "Nonlinear aspects of sampled-data control systems," in *Proc. of Symp. on Nonlinear Circuit Analysis* (J. Fox, ed.), Polytechnic Institute of Brooklyn, 1956, pp. 273-313.
214. Kalman, R.E., "A new approach to linear filtering and prediction problems," *ASME Trans.* **82D**(1960): 33-45.
215. Kalman, R.E., "Contributions to the theory of optimal control," *Bol. Soc. Mat. Mex.* **5**(1960): 102-119.

216. Kalman, R.E., "On the general theory of control systems," in *Proc. First IFAC Congress Automatic Control*, Moscow, 1960, Butterworths, London, vol. 1, pp. 481-492.
217. Kalman, R.E., "Canonical structure of linear dynamical systems," *Proc. Natl. Acad. Sci.* **48**(1962): 596-600.
218. Kalman, R.E., "Mathematical description of linear dynamical systems," *SIAM J. Control* **1**(1963): 152-192.
219. Kalman, R.E., "The theory of optimal control and the calculus of variations," in *Mathematical Optimization Techniques* (R. Bellman, ed.), Univ. of California Press, 1963, pp. 309-331.
220. Kalman, R.E., "Algebraic structure of linear dynamical systems, I. The module of Σ ," *Proc. Natl. Acad. Sci. (USA)* **54**(1965): 1503-1508.
221. Kalman, R.E., "Irreducible realizations and the degree of a rational matrix," *J. SIAM* **13**(1965): 520-544.
222. Kalman, R.E., "Linear stochastic filtering - Reappraisal and outlook," in *Proc. Symp. on System Theory* (J. Fox, ed.), Polytechnic Press, New York, 1965, pp. 197-205.
223. Kalman, R.E., "Algebraic aspects of the theory of dynamical systems," in *Differential Equations and Dynamical Systems* (J.K. Hale and J.P. LaSalle, eds.), Academic Press, New York, 1967, pp. 133-146.
224. Kalman, R.E., "Kronecker invariants and feedback," in *Ordinary Differential Equations* (L. Weiss, ed.), Academic Press, New York, 1972, pp. 459-471.
225. Kalman, R.E., "Algebraic-geometric description of the class of linear systems of constant dimension," in *Proc. 8th Princeton Conf. Inform. Sci. and Systems*, 1974, pp. 189-191.
226. Kalman, R.E., "Optimization, mathematical theory of, control theory" in *Encyclopaedia Britannica, Fifteenth Edition*, 1974, pp. 636-638.
227. Kalman, R.E., "Algebraic aspects of the generalized inverse," in *Generalized Inverses and Applications* (M. Zuhair Nashed, ed.), Academic Press, New York, 1976, pp. 111-124.
228. Kalman, R.E., "Identifiability and modeling in econometrics," *Developments in Statistics* **4**(1983): 97-135.
229. Kalman, R.E., and R.S. Bucy, "New results in linear filtering and prediction theory," *Trans. ASME (J. Basic Eng.)* **83D**(1961): 95-108.
230. Kalman, R.E., P.L. Falb, and M.A. Arbib, *Topics in Mathematical System Theory*, McGraw-Hill, New York, 1969.
231. Kalman, R.E., Y.C. Ho, and K.S. Narendra, "Controllability of linear dynamical systems," *Control Diff. Eqs.* **1**(1963): 189-213.
232. Kalouptsidis, N., and J. Tsinias, "Stability improvement of nonlinear systems by feedback," *IEEE Trans. Autom. Control* **29**(1984): 346-367.
233. Kamen, E.W., "On an algebraic theory of systems defined by convolution operators," *Math. Systems Theory* **9** (1975): 57-74.

- 234. Kamen, E.W., "Module structure of infinite-dimensional systems with applications to controllability," *SIAM J. Control & Opt.* **14**(1976):389-408.
- 235. Kamen, E.W., P.K. Khargonekar, and K.R. Polla, "A transfer function approach to linear time-varying systems," *SIAM J. Control & Opt.* **23**(1985): 550-565.
- 236. Kappel, F., K. Kunisch, and W. Schappacher, *Control Theory for Distributed Parameter Systems and Applications*, Springer-Verlag, New York, 1983.
- 237. Karmy, M., K. Warwick, and V. Kurkova (eds.), *Dealing with Complexity: a Neural Network Approach*, Springer-Verlag, London, 1997.
- 238. Kawski, M., "Stabilization of nonlinear systems in the plane," *Systems and Control Letters* **12**(1989): 169-176.
- 239. Kawski, M., "High-order small-time local controllability," in *Nonlinear Controllability and Optimal Control* (H.J. Sussmann, ed.), Dekker, New York, 1990, pp. 431-467.
- 240. Kawski, M., "The complexity of deciding controllability," in *Systems Control Lett.* **15** (1990): 9-14.
- 241. Kawski, M., and Sussmann, H.J., "Noncommutative power series and formal Lie-algebraic techniques in nonlinear control theory," in *Operators, Systems and Linear Algebra: Three Decades of Algebraic Systems Theory* (U. Helmke, D. Praetzel-Wolters, and E. Zerz, eds.), B.G. Teubner, Stuttgart, 1997, pp. 111-129.
- 242. Kenney, C., and A.J. Laub, "Controllability and stability radii for companion form systems," *Math. Control Signals Systems* **1**(1988):239-256.
- 243. Khargonekar, P.P., "On matrix fraction representations for linear systems over commutative rings," *SIAM J. Control & Opt.* **20**(1982): 172-197.
- 244. Khargonekar, P.P., T.T. Georgiou, and A.M. Pascoal, "On the robust stabilizability of linear time invariant plants with unstructured uncertainty," *IEEE Trans. Autom. Control* **AC-32**(1987): 201-208.
- 245. Khargonekar, P.P., K. R. Poolla, and A. Tannenbaum, "Robust control of linear time-invariant plants by periodic compensation," *IEEE Trans. Autom. Control* **AC-30**(1985): 1088-1096.
- 246. Khargonekar, P.P., and E.D. Sontag, "On the relation between stable matrix fraction decompositions and regulable realizations of systems over rings," *IEEE Trans. Autom. Control* **27**(1982): 627-638.
- 247. Kirk, D.E., *Optimal Control Theory; an Introduction*, Prentice-Hall, Englewood Cliffs, 1970.
- 248. Knobloch, H.W., *Higher Order Necessary Conditions in Optimal Control Theory*, Springer-Verlag, New York, 1981.
- 249. Koditschek, D.E., "Adaptive techniques for mechanical systems," in *Proc. 5th. Yale Workshop on Adaptive Systems*, Yale University, New Haven, 1987, pp. 259-265.

250. Kokotovic, P.V., and H.J. Sussmann, "A positive real condition for global stabilization of nonlinear systems," *Systems Control Letters* **13**(1989): 125-134.
251. Kolmanovsky, I., and N.H. McClamroch, "Developments in nonholonomic control problems," *Control Systems Magazine* **15**(1995): 20-36.
252. Koppel, L.B., *Introduction to Control Theory, with Applications to Process Control*, Prentice-Hall, Englewood Cliffs, 1968.
253. Kou, S.R., D.L. Elliott, and T.J. Tarn, "Exponential observers for nonlinear dynamical systems," *Inform. Contr* **29**(1975): 204-216.
254. Krasnoselskii, M.A., and P.P. Zabreiko, *Geometric Methods of Nonlinear Analysis*, Springer-Verlag, Berlin, 1983.
255. Krener, A.J., "A generalization of Chow's theorem and the bang-bang theorem to nonlinear control systems," *SIAM J. Control* **12** (1974): 43-52.
256. Krener, A.J., "The high order maximal principle and its application to singular extremals," *SIAM J. Control & Opt.* **15**(1977): 256-293.
257. Krener, A.J., "(Adf,g), (adf,g) and locally (adf,g) Invariant and Controllability Distributions," *SIAM J. Control & Opt.* **23**(1985): 523-549.
258. Krener, A.J., "Nonlinear controller design via approximate normal forms," in *Proc. IMA Conf. on Signal Processing, Minneapolis, June-Aug. 1988*, Institute for Mathematics and its Applications, 1989.
259. Krstic, M., I. Kanellakopoulos, and P. Kokotovic, *Nonlinear and Adaptive Control Design*, Wiley, New York, 1995.
260. Kuo, B.C., *Automatic Control Systems*, Seventh Edition, Prentice Hall, Upper Saddle River, NJ, 1995.
261. Kupka, I., and G. Sallet, "A sufficient condition for the transitivity of pseudo-groups: Application to system theory," *J. Diff. Eqs.* **47**(1973): 462-470.
262. Kurzweil, J., "On the inversion of Lyapunov's second theorem on stability of motion," *Amer. Math. Soc. Transl., Ser.2*, **24** (1956): 19-77.
263. Kwakernaak, H., and R. Sivan, *Linear Optimal Control Systems*, Wiley, New York, 1972.
264. Lang, S., *Analysis II*, Addison-Wesley, Reading, MA, 1969.
265. Langenhop, C.E., "On the stabilization of linear systems," *Proc. AMS* **15**(1964): 735-742.
266. Lee, E.B., and L. Markus, *Foundations of Optimal Control Theory*, Wiley, New York, 1968.
267. Lee, K.K., and A. Arapostathis, "On the controllability of piecewise linear hypersurface systems," *Systems & Control Letters* **9**(1987): 89-96.
268. Lee, K.K., and A. Arapostathis, "Remarks on smooth feedback stabilization of nonlinear systems," *Systems & Control Letters* **10**(1988):41-44.
269. Lee, H.G., A. Arapostathis, and S.I. Marcus, "On the linearization of discrete time systems," *Int. J. Control* **45**(1987): 1103-1124.

- 270. Leigh, J.R., *Functional Analysis and Linear Control Theory*, Academic Press, New York, 1980.
- 271. Lenz, K., J. Doyle, and P. Khargonekar, "When is a controller H_∞ optimal?" *Math. Control Signals Systems* **1**(1988): 107-122.
- 272. Leontaritis, I.J., and S.A. Billings, "Input-output parametric models for nonlinear systems," Parts I and II, *Int. J. Control* **41** (1985): 303-344.
- 273. Levi, M., and F.M.A. Salam (eds.), *Dynamical Systems Approaches to Nonlinear Problems in Systems and Circuits*, SIAM Publications, Philadelphia, 1988.
- 274. Levine, W.S. (ed.), *The Control Handbook* CRC Press, Boca Raton, 1995, pp. 895-908.
- 275. Levis, A.H., et. al., "Challenges to control: A collective view. Report from the workshop held at the University of Santa Clara on September 18-19, 1986," *IEEE Trans. Autom. Control* **32**(1987): 275-285.
- 276. Lewis, F.L., *Optimal Estimation: With an Introduction to Stochastic Control Theory*, Wiley, New York, 1986.
- 277. Lewis, F.L., "A survey of linear singular systems," *Circuits, Sys., and Signal Proc.* **5**(1986): 3-36.
- 278. Lin, Y., and E.D. Sontag, "Control-Lyapunov universal formulae for restricted inputs," *Control: Theory and Advanced Technology* **10**(1995): 1981-2004.
- 279. Lin, Y., E.D. Sontag, and Y. Wang, "A smooth converse Lyapunov theorem for robust stability," *SIAM J. Control and Optimization* **34** (1996): 124-160.
- 280. Lindquist, A., and G. Picci, "Realization theory for multivariate stationary Gaussian processes," *SIAM J. Control & Opt.* **23**(1985): 809-857.
- 281. Lions, J.L., *Optimal Control of Systems Governed by Partial Differential Equations*, Springer-Verlag, Berlin, 1971.
- 282. Ljung, L., and T. Söderström, *Theory and Practice of Recursive Estimation*, MIT Press, Cambridge, 1983.
- 283. Lo, J.T., "Global bilinearization of systems with controls appearing linearly," *SIAM J. Control* **13**(1975): 879-885.
- 284. Lobry, C., "Contrôlabilité des systèmes non linéaires," *SIAM J. Control* **8**(1970): 573-605.
- 285. Logemann, H., "On the transfer matrix of a neutral system: Characterizations of exponential stability in input-output terms," *Systems and Control Letters* **9**(1987): 393-400.
- 286. Longchamp, R., "State-feedback control of bilinear systems," *IEEE Trans. Autom. Contr.* **25**(1980): 302-306.
- 287. Luenberger, D.G., "An introduction to observers," *IEEE Trans. Autom. Contr.* **16**(1971): 596-602.
- 288. MacFarlane, A.G.J., "An eigenvector solution of the optimal linear regulator," *J. Electron. Control* **14**(1963): 643-654.

289. Macki, J., and A. Strauss., *Introduction to Optimal Control Theory*, Springer-Verlag, New York, 1982.
290. Manes, E.G., ed., *Category Theory Applied to Computation and Control, Proceedings of the First International Symposium* (San Francisco, February 25-26, 1974), Springer-Verlag, New York, 1975.
291. Manitius, A., "Necessary and sufficient conditions of approximate controllability for general linear retarded systems," *SIAM J. Control & Opt.* **19**(1981): 516-532.
292. Marcus, S.I., "Optimal nonlinear estimation for a class of discrete time stochastic systems," *IEEE Trans. Autom. Contr.* **24**(1979): 297-302.
293. Marcus, S.I., "Algebraic and geometric methods in nonlinear filtering," *SIAM J. Control & Opt.* **22**(1984): 817-844.
294. Marino, R., "Feedback stabilization of single-input nonlinear systems," *Systems and Control Letters* **10**(1988): 201-206.
295. Markus, L., and H. Yamabe, "Global stability criteria for differential systems," *Osaka J. Math.* **12**(1960): 305-317.
296. Massera, J.L., "Contributions to stability theory," *Annals of Math* **64** (1956): 182-206. Erratum in *Annals of Math* **68**(1958): 202.
297. Massey, J.L., and M.K. Sain, "Codes, automata and continuous systems: Explicit interconnections," *IEEE Trans. Autom. Control* **12**(1970): 644-650.
298. Mayeda, H., and T. Yamada, "Strong structural controllability," *SIAM J. Control & Opt.* **17**(1979): 123-138.
299. Mayr, O., *The Origins of Feedback Control*, MIT Press, Cambridge, MA, 1970.
300. Mesarovic, M.D., and Y. Takahara, *General Systems Theory: Mathematical Foundations*, Academic Press, New York, 1975.
301. Minorsky, N., *Theory of Nonlinear Control Systems*, McGraw-Hill, New York, 1969.
302. Monaco, S., and D. Normand-Cyrot, "Invariant distributions for nonlinear discrete-time systems," *Systems and Control Letters* **5**(1984): 191-196.
303. Moog, C. H., "Nonlinear decoupling and structure at infinity," *Math. Control Signals Systems* **1**(1988): 257-268
304. Moroney, P., *Issues in the Implementation of Digital Feedback Compensators*, MIT Press, Cambridge, MA, 1983.
305. Morse, A.S., "Ring models for delay differential systems," *Automatica* **12**(1976): 529-531.
306. Morse, A.S., and L.M. Silverman, "Structure of index-invariant systems," *SIAM J. Control* **11**(1972): 215-225.
307. Narendra, K.S., and K. Parthasarathy, "Identification and control of dynamical systems using neural networks," *IEEE Trans. Neural Nets* **1**(1990): 4-27.
308. Nelson, E., *Tensor Analysis*, Princeton University Press, Princeton, NJ, 1967.

309. Nieuwenhuis, J.W., and J.C. Willems, "Continuity of dynamical systems - a system theoretic approach," *Math. Control Signals Systems* **1**(1988): 147-165.
310. Nijmeijer, H., "Observability of autonomous discrete-time nonlinear systems: a geometric approach," *Int. J. Control* **36**(1982): 867-874.
311. Nijmeijer, H., and A.V. Van der Schaft, *Nonlinear Dynamical Control Systems*, Springer-Verlag, New York, 1990.
312. Ocone, O., "Probability densities for conditional statistics in the cubic sensor problem," *Math. Control Signals Systems* **1**(1988): 183-202.
313. Ozveren, C.M., G.C. Verghese, and A.S. Willsky, "Asymptotic orders of reachability in perturbed linear systems," *IEEE Trans. Autom. Control* **33**(1988): 915-923.
314. Padulo, L., and M.A. Arbib, *System Theory. A Unified State-Space Approach to Continuous and Discrete Systems*, Saunders, Philadelphia, 1974. Republished by Hemisphere, Washington, DC.
315. Pallu de La Barriere, R., *Optimal Control Theory; A Course in Automatic Control Theory*, Saunders, Philadelphia, 1967.
316. Pandolfi, L., "Canonical realizations of systems with delays," *SIAM J. Control & Opt.* **21**(1983): 598-613.
317. Pitchford, J.D., and S.J. Turnovsky, *Applications of Control Theory to Economic Analysis*, Elsevier/North-Holland, Amsterdam, 1977.
318. Polderman, J.W., "A state space approach to the problem of adaptive pole placement," *Math. Control Signals Systems* **2**(1989): 71-94.
319. Pontryagin, L.S., V.G. Boltyanskii, R.V. Gamkrelidze, and E.F. Mishchenko, *The Mathematical Theory of Optimal Processes*, Wiley, New York, 1962.
320. Popov, V.M., "On a new problem of stability for control systems," *Autom. Remote Control* (English translation) **24**(1963): 1-23.
321. Popov, V.M., *Hyperstability of Control Systems*, Springer-Verlag, Berlin, 1973. (Translation of *Hiperstabilitatea Sistemelor Automate*, published in 1966 in Rumanian.)
322. Potter, J.E., "Matrix quadratic solutions," *SIAM J. Appl. Math.* **14** (1966): 496-501.
323. Praly, L., and Y. Wang, "Stabilization in spite of matched unmodelled dynamics and an equivalent definition of input-to-state stability," *Math. Control Signals Systems* **9**(1996): 1-33.
324. Prichard, A.J., and J. Zabczyk, "Stability and stabilizability of infinite dimensional systems," *SIAM Review* **23**(1983): 25-52.
325. Ramirez, W.F., *Application of Optimal Control Theory to Enhanced Oil Recovery*, Elsevier, New York, 1987.
326. Rantala, J., *An Application of Stochastic Control Theory to Insurance Business*, University of Tampere, Tampere, 1984.
327. Reutenauer, C., *Free Lie Algebras*, Oxford University Press, New York, 1993.

- 328. Riggs, D.S., *Control Theory and Physiological Feedback Mechanisms*, Krieger, Huntington, New York, 1976.
- 329. Rissanen, J., "Control system synthesis by analogue computer based on the 'generalized linear feedback' concept," in *Proc. Symp. Analoge Comp. Applied to the Study of Chem. Processes*, Presses Académiques Européennes, Brussels, 1961.
- 330. Rockafellar, R.T., *Convex Analysis*, Princeton University Press, 1970.
- 331. Rosenbrock, H.H., *State Space and Multivariable Theory*, Nelson, London, 1970.
- 332. Rouchaleau, Y., B.F. Wyman, and R.E. Kalman, "Algebraic structures of linear dynamical systems III. Realization theory over a commutative ring," *Proc. Nat. Acad. Sci.* **69**(1972): 3404-3406.
- 333. Rouchaleau, Y., and B.F. Wyman, "Linear dynamical systems over integral domains," *J. Comput. System Sci.* **9**(1974): 129-142.
- 334. Rouche, N., P. Habets, and M. Laloy, *Stability Theory by Lyapunov's Direct Method*, Springer-Verlag, New York, 1977.
- 335. Rugh, W.J., *Mathematical Description of Linear Systems*, Marcel Dekker, New York, 1975.
- 336. Rugh, W.J., *Nonlinear System Theory: the Volterra/Wiener Approach*, The Johns Hopkins University Press, Baltimore, 1981.
- 337. Russell, D.L., *Mathematics of Finite Dimensional Control Systems*, Marcel Dekker, New York, 1979.
- 338. Ryan, E.P., "On Brockett's condition for smooth stabilizability and its necessity in a context of nonsmooth feedback," *SIAM J. Control Optim.* **32**(1994): 1597-1604.
- 339. Ryan, E.P., and N.J. Buckingham, "On asymptotically stabilizing feedback control of bilinear systems," *IEEE Trans. Autom. Contr.* **28**(1983): 863-864.
- 340. Sage, A.P., and J.L. Melsa, *Estimation Theory with Applications to Communications and Control*, Krieger, Huntington, 1979.
- 341. Salam, F.M.A., "Feedback stabilization of the nonlinear pendulum under uncertainty: a robustness issue," *Systems and Control Letters* **7**(1986): 199-206.
- 342. Saskaena, V.R., J.O'Reilly, and P.V. Kokotovic, "Singular perturbations and two-scale methods in control theory: survey 1976-1983," *Automatica* **20**(1984): 273-293.
- 343. Schmitendorf, W.E., and B.R. Barmish, "Null controllability of linear systems with constrained controls," *SIAM J. Control & Opt.* **18**(1980): 327-345.
- 344. Schwartz, C.A., B.W. Dickinson, and E.D. Sontag, "Characterizing innovation representations for discrete-time random processes," *Stochastics* **11**(1984): 159-172.
- 345. Seidman, T.I., "How violent are fast controls?," *Math. Control Signals Systems* **1**(1988): 89-95.

- 346. Seierstad, A., and K. Sydsater, *Optimal Control Theory with Economic Applications*, North-Holland, New York, 1987.
- 347. Sethi, S.P., and G.L. Thompson, *Optimal Control Theory: Applications to Management Science*, Kluwer, Boston, 1981.
- 348. Shaefer, H.H., *Topological Vector Spaces*, Springer-Verlag, New York, 1980.
- 349. Sharma, P.K., "On pole assignment problem in polynomial rings," *Systems and Control Letters* **5**(1984): 49-54.
- 350. Siljak, D.D., *Large Scale Dynamic Systems*, North-Holland, New York, 1978.
- 351. Silva Leite, F., and P.E. Crouch, "Controllability on classical Lie groups," *Math. Control Signals Systems* **1**(1988): 31-42.
- 352. Singh, M.G. (ed.), *Systems & Control Encyclopedia: Theory, Technology, Applications*, Pergamon Press, Oxford, 1987.
- 353. Sira-Ramirez, H., "A geometric approach to pulse-width-modulated control in dynamical systems," *IEEE Trans. Autom. Control* **34** (1989): 184-187.
- 354. Skowronski, J.M., *Control Theory of Robotic Systems*, World Scientific, Teaneck, 1989.
- 355. Slemrod, M., "Stabilization of bilinear control systems with applications to nonconservative problems in elasticity," *SIAM J. Control & Opt.* **16**(1978): 131-141.
- 356. Sontag, E.D., "On certain questions of rationality and decidability," *J. Comp. Syst. Sci.* **11**(1975): 375-381.
- 357. Sontag, E.D., "Linear systems over commutative rings: A survey," *Ricerche di Automatica* **7**(1976): 1-34.
- 358. Sontag, E.D., "On the observability of polynomial systems," *SIAM J. Control & Opt.* **17**(1979):139-151.
- 359. Sontag, E.D., "Realization theory of discrete-time nonlinear systems: Part I- The bounded case," *IEEE Trans. Circuits and Syst.* **CAS-26**(1979): 342-356.
- 360. Sontag, E.D., *Polynomial Response Maps*, Springer-Verlag, New York, 1979.
- 361. Sontag, E.D., "Conditions for abstract nonlinear regulation," *Information and Control* **51**(1981): 105-127.
- 362. Sontag, E.D., "Nonlinear regulation: The piecewise linear approach," *IEEE Trans. Autom. Control* **AC-26**(1981): 346-358.
- 363. Sontag, E.D., "A Lyapunov-like characterization of asymptotic controllability," *SIAM J. Control & Opt.* **21**(1983): 462-471.
- 364. Sontag, E.D., "An algebraic approach to bounded controllability of nonlinear systems," *Int. J. Control* **39**(1984): 181-188.
- 365. Sontag, E.D., "A concept of local observability," *Systems and Control Letters* **5**(1984): 41-47.

366. Sontag, E.D., "Controllability is harder to decide than accessibility," *SIAM J. Control & Opt.* **26**(1988): 1106-1118.
367. Sontag, E.D., "Finite dimensional open-loop control generators for nonlinear systems," *Int. J. Control* **47** (1988): 537-556.
368. Sontag, E.D., "Remarks on stabilization and input-to-state stability," *Proc. IEEE Conf. Dec. and Control, Tampa, Dec. 1989*, IEEE Publications, pp. 1376-1378.
369. Sontag, E.D., "A 'universal' construction of Artstein's theorem on nonlinear stabilization," *Systems and Control Letters* **13**(1989): 117-123.
370. Sontag, E.D., "Smooth stabilization implies coprime factorization," *IEEE Trans. Autom. Control* **34**(1989): 435-443.
371. Sontag, E.D., "Integrability of certain distributions associated with actions on manifolds and applications to control problems," in *Nonlinear Controllability and Optimal Control* (H.J. Sussmann, ed.), pp. 81-131, Marcel Dekker, New York, 1990.
372. Sontag, E.D., "Feedback stabilization of nonlinear systems," in *Robust Control of Linear Systems and Nonlinear Control* (M.A. Kaashoek, J.H. van Schuppen, and A.C.M. Ran, eds.), Birkhäuser, Cambridge, MA, 1990, pp. 61-81.
373. Sontag, E.D., "Spaces of observables in nonlinear control," in *Proc. Intern. Congress of Mathematicians 1994*, Volume 2, Birkhäuser Verlag, Basel, 1995, pp. 1532-1545.
374. Sontag, E.D., "Control of systems without drift via generic loops," *IEEE Trans. Autom. Control* **40**(1995): 1210-1219.
375. Sontag, E.D., and H.J. Sussmann, "Complete controllability of continuous-time recurrent neural networks," *Systems and Control Letters* **30**(1997): 177-183.
376. Sontag, E.D., and Y. Wang, "New characterizations of the input to state stability property," *IEEE Trans. Autom. Control* **AC-41**(1996): 1283-1294.
377. Sontag, E.D., and Y. Wang, "Output-to-state stability and detectability of nonlinear systems," *Systems and Control Letters* **29**(1997): 279-290.
378. Spong, M.W., and M. Vidyasagar, *Robot Dynamics and Control*, Wiley, New York, 1989.
379. Stengel, R.F., *Stochastic Optimal Control: Theory and Application*, Wiley, New York, 1986.
380. Stephanopoulos, G., *Chemical Process Control: An Introduction to Theory and Practice*, Prentice-Hall, Englewood Cliffs, 1984.
381. Sussmann, H.J., "Semigroup representations, bilinear approximations of input-output maps, and generalized inputs," in *Mathematical Systems Theory, Udine 1975* (G. Marchesini, ed.), Springer-Verlag, New York, pp. 172-192.
382. Sussmann, H.J., "Minimal realizations and canonical forms for bilinear systems," *J. Franklin Institute* **301**(1976): 593-604.

- 383. Sussmann, H.J., "Existence and uniqueness of minimal realizations of non-linear systems," *Math. Sys. Theory***10** (1977): 263-284.
- 384. Sussmann, H.J., "Single-input observability of continuous-time systems," *Math. Systems Theory* **12**(1979): 371-393.
- 385. Sussmann, H.J., "Subanalytic sets and feedback control," *J.Diff. Eqs.* **31**(1979): 31-52.
- 386. Sussmann, H.J., "Analytic stratifications and control theory," in *Proc. 1978 Internat. Congr. Mathematicians, Helsinki*, pp. 865-871.
- 387. Sussmann, H.J., "Lie brackets, real analyticity, and geometric control," in *Differential Geometric Control theory* (R.W. Brockett, R.S. Millman, and H.J. Sussmann, eds.), Birkhauser, Boston, 1983.
- 388. Sussmann, H.J., "A general theorem on local controllability," *SIAM J. Control & Opt.* **25**(1987): 158-194.
- 389. Sussmann, H.J., "Minimal realizations of nonlinear systems," in *Geometric Methods in Systems Theory* (D.Q. Mayne and R.W. Brockett, eds.), Reidel, Dordrecht, 1973, pp. 243-252.
- 390. Sussmann, H.J., "Multidifferential calculus: chain rule, open mapping and transversal intersection theorems," in *Optimal Control: Theory, Algorithms, and Applications* (W.W. Hager and P.M. Pardalos, eds.), Kluwer, 1997.
- 391. Sussmann, H.J., and V. Jurdjevic, "Controllability of nonlinear systems," *J. Diff. Eqs.* **12**(1972): 95-116.
- 392. Sussmann, H.J., E.D. Sontag, and Y. Yang, "A general result on the stabilization of linear systems using bounded controls," *IEEE Trans. Autom. Control* **39**(1994): 2411-2425.
- 393. Sussmann, H.J., and J.C. Willems, "300 years of optimal control: From the brachystochrone to the maximum principle," *Control Systems Magazine* **17**(1997): 32-44.
- 394. Swan, G.W., *Applications of Optimal Control Theory in Biomedicine*, Marcel Dekker, New York, 1984.
- 395. Tannenbaum, A., *Invariance and System Theory: Algebraic and Geometric Aspects*, Springer-Verlag, New York, 1980.
- 396. Tannenbaum, A., "On pole assignability over polynomial rings," *Systems and Control Letters* **2**(1982):13-16.
- 397. Tannenbaum, A., "Polynomial rings over arbitrary fields in two or more variables are not pole assignable," *Systems and Control Letters* **2**(1982): 222-224.
- 398. Tay, T.T., and J.B. Moore, "Left coprime factorizations and a class of stabilizing controllers for nonlinear systems," in *Proc. IEEE Conf. Decision and Control*, Austin, Dec.1988, pp. 449-454.
- 399. Taylor, A.E., *Introduction to Functional Analysis*, Wiley, New York, 1958.

400. Thompson, W. (Lord Kelvin), "Mechanical integration of the general linear differential equation of any order with variable coefficients," *Proc. Roy. Soc.* **24**(1876): 271-275.
401. Toates, F.M., *Control Theory in Biology and Experimental Psychology*, Hutchinson, London, 1975.
402. Troutman, J.L., *Variational Calculus and Elementary Convexity*, Springer Verlag, New York, 1983.
403. Tsiniias, J., "Sufficient Lyapunovlike conditions for stabilization," *Math. Control Signals Systems* **2**(1989): 343-357.
404. Tsiniias, J., and N. Kalouptsidis, "Prolongations and stability analysis via Lyapunov functions of dynamical polysystems," *Math. Systems Theory* **20**(1987): 215-233.
405. Tsitsiklis, J.N., "On the control of discrete-event dynamical systems," *Math. Control Signals Systems* **2**(1989): 95-107.
406. Utkin, V.I., *Sliding Modes and Their Application in Variable Structure Systems*, Mir, Moscow, 1978.
407. Van der Schaft, A.J., "Stabilization of Hamiltonian Systems," *J. Nonlinear Analysis, MTA* **10**(1986): 1021-1035.
408. Van der Schaft, A.J., "On realizations of nonlinear systems described by higher-order differential equations," *Math Systems Theory* **19** (1987): 239-275.
409. Van Schuppen, J.H., "Stochastic realization problems," in *Three Decades of Mathematical System Theory* (Nijmeijer, H., and J.M. Schumacher, eds.), Springer-Verlag, Berlin, 1989.
410. Varaiya, P.P., and R. Liu, "Bounded-input bounded-output stability of nonlinear time-varying differential systems," *SIAM J. Control* **4**(1966): 698-704.
411. Veliov, V.M., and M.I. Krastanov, "Controllability of piecewise linear systems," *Systems and Control Letters* **7**(1986): 335-341.
412. Verma, M.S., "Coprime fractional representations and stability of nonlinear feedback systems," *Int. J. Control* **48**(1988): 897-918.
413. Vidyasagar, M., "Input-output stability of a broad class of linear time-invariant multivariable systems," *SIAM J. Ctr.* **10**(1972):203-209.
414. Vidyasagar, M., *Nonlinear Systems Analysis*, Prentice-Hall, Englewood Cliffs, 1978.
415. Vidyasagar, M., "Decomposition techniques for large-scale systems with nonadditive interactions: stability and stabilizability," *IEEE Trans. Autom. Control* **25** (1980): 773-779.
416. Vidyasagar, M., "On the stabilization of nonlinear systems using state detection," *IEEE Trans. Autom. Control* **25**(1980): 504-509.
417. Vidyasagar, M., *Control Systems Synthesis: A Factorization Approach*, MIT Press, Cambridge, MA, 1985.

- 418. Vinter, R.B., "New results on the relationship between dynamic programming and the maximum principle," *Math. Control Signals Systems* **1**(1988): 97-105.
- 419. Viswanadham, N., and M. Vidyasagar, "Stabilization of linear and non-linear dynamical systems using an observer-controller configuration," *Systems and Control Letters* **1** (1981): 87-91.
- 420. Vukobratovic, M., and D. Stokic, *Control of Manipulation Robots: Theory and Application*, Springer-Verlag, Berlin, 1982.
- 421. Wang, Y., and E.D. Sontag, "Algebraic differential equations and rational control systems," *SIAM J. Control and Opt.* **30**(1992): 1126-1149.
- 422. Willems, J.C., *The Analysis of Feedback Systems*, MIT Press, Cambridge, MA, 1971.
- 423. Willems, J.C., "Least squares stationary optimal control and the algebraic Riccati equation," *IEEE Trans. Autom. Control* **AC-16**(1971): 621-634.
- 424. Willems, J.C., "From time series to linear systems, Part I: Finite dimensional time-invariant systems," *Automatica* **22**(1986):561-580.
- 425. Willems, J.C., "From time series to linear systems, Part II: Exact Modelling," *Automatica* **22**(1986): 675-694.
- 426. Willems, J.C., "From time series to linear systems, Part III: Approximate Modelling," *Automatica* **23**(1987): 87-115.
- 427. Willsky, A.S., *Digital Signal Processing and Control and Estimation Theory: Points of Tangency, Areas of Intersection, and Parallel Directions*, MIT Press, Cambridge, MA, 1979.
- 428. Wilson, F.W., "Smoothing derivatives of functions and applications," *Trans. AMS* **139**(1969): 413-428.
- 429. Windeknecht, T.G., *General Dynamical Processes*, Academic Press, New York, 1971.
- 430. Wonham, M., *Linear Multivariable Control*, Springer-Verlag, New York, 1974.
- 431. Wonham, W.M., "On pole assignment in multi-input controllable linear systems," *IEEE Trans. Autom. Control* **AC-12**(1967):660-665.
- 432. Wonham, W.M., and A.S. Morse, "Decoupling and pole assignment in linear multivariable systems: a geometric approach," *SIAM J. Control* **8**(1970): 1-18.
- 433. Wonham, W.M., and P.J. Ramadge, "Modular supervisory control of discrete event systems," *Math. Control Signals Systems* **1**(1988): 13-30.
- 434. Wyman, B.F., "Time-varying linear discrete-time systems - realization theory," in *Advances in Math., Supp. Studies, Vol. I*, Academic Press, New York, 1978, 233-258.
- 435. Wyman, B.F., "Time-varying linear discrete-time systems: duality," *Pacific J. of Math.* **86**(1980): 361-377.
- 436. Wyman, B.F., and M.K. Sain, "The zero module and essential inverse systems," *IEEE Trans. Circuits and Systems* **CAS-28**(1981): 112-126.

- 437. Yamamoto, Y., "Realization theory of infinite dimensional linear systems, Parts I,II," *Math Sys. Theory* **15**(1981): 55-77, 169-190.
- 438. Yamamoto, Y., "Pseudo-rational input/output maps and their realizations: a fractional representation approach to infinite-dimensional systems," *SIAM J. Control & Opt.* **26-6**(1988): 1415-1430.
- 439. Yamamoto, Y., "Correspondence of internal and external stability - Realization, transfer functions and complex analysis," in *Realization and Modelling in System Theory* (M.A. Kaashoek, J.H. van Schuppen, and A.C.M. Ran, eds.), Birkhäuser, Cambridge, MA, 1990.
- 440. Yamamoto, Y., and E.D. Sontag, "On the existence of approximately coprime factorizations for retarded systems," *Systems and Control Letters* **13**(1989): 53-58.
- 441. Youla, D.C., H.A. Jabr, and J.J. Bongiorno, "Modern Wiener-Hopf design of optimal controllers: part II," *IEEE Trans. Autom. Contr.* **AC-21**(1976): 319-338.
- 442. Young, L.C., *Lectures on the Calculus of Variations and Optimal Control Theory*, Chelsea, New York, 1980.
- 443. Zabczyk, J., "Some comments on stabilizability," *Applied Math Optimiz.* **19**(1989): 1-9.
- 444. Zadeh, L.A., and C.A. Desoer, *Linear System Theory — The State-Space Approach*, McGraw-Hill, New York, 1963.